Chapter 1

Explainability in Reinforcement Learning

Today, the majority of artificial intelligence agents with which humans interact are deployed in relatively controlled and low-risk environments. These agents are currently mainly used in simulated, digital, or non-critical impact settings, such as games, recommendation systems, online advertising, or computer system optimization. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences. Even more autonomous systems, such as agents based on large language models, remain strongly constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval. Safety-critical applications integrating machine learning components are still relatively rare.

However, many potential high-impact applications such as drone control, autonomous delivery, autonomous vehicle driving, adaptive urban traffic management, optimization of smart power grids, or control of industrial robotic systems in real-world environments would require significantly higher levels of autonomy. Introducing autonomous agents in such environments, requires the establishment of strong guarantees. These include in particular:

- validating the system's behavior before deployment,
- being able to identify the origin of a malfunction in case of failure,
- and being able to certify that an observed error will not occur again.

Without satisfying these criteria, deploying autonomous agents in the real world remains hardly feasible.

The main challenge lies in the fact that the most performant agents currently available rely on reinforcement learning methods. This learning process is carried out without explicit human supervision, and these agents are not able to provide explanations for the mechanisms that lead to their decisions. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, a research field has emerged: Explainable Reinforcement Learning (XRL), a subfield of Explainable Artificial Intelligence (XAI).

Although relatively recent, this field is rapidly growing and offers a wide variety of approaches. This chapter aims to present the main existing XRL techniques. To this end, we first define what is meant by an “explanation” in the context of reinforcement learning. We then propose a structuring of the XRL research field. Finally, we review the state of the art following this organization.

1.1 Explanation

It is necessary to establish what is meant by an explanation in the context of XAI. The difficulty of this question lies in its fundamentally multidisciplinary nature. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI. For this reason, we introduce here a working definition of explanation.

An explanation involves two entities: the sujet being explained and the receiver of the explanation. The receiver of the explanation is a human, who will be referred to as the observer throughout this manuscript. The observer aims to understand how the subject works and, in doing so, builds a mental model of its functioning. In this context, an explanation is what allows the observer to refine and correct their mental model of how the system operates.

Definition 1.1.1: Subject

The subject of an explanation is the system is to be understood by an observer.

Definition 1.1.2: Observer

The observer is the human. The observer possesses a mental model of the subject of the explanation, an internal representation of how they believe the system operates. This mental model may be incomplete or incorrect.

Definition 1.1.3: Explanation

An explanation is any information that enables an observer to refine or correct their mental model of the subject of the explanation.

We can already see that this definition of explanation is deliberately broad. This is intentional, as our goal is not to engage in sterile debates over definitions. It also allows us to nuance the notion of explanation by considering the effectiveness of an explanation. Intuitively, providing information about a very rare aspect of the system does not have the same value as correcting the observer’s misunderstanding of a fundamental mechanism of the system.

Now that a general definition of explanation has been established, we can focus on its meaning in the context of XAI. In this context, the subject of the explanation is an artificial intelligence system. More precisely, XAI aims at explaining deep neural network models. The fact that deep learning models encode knowledge through connectionist representations makes it impossible to understand their functioning simply by inspecting the model. As a result, they are universally regarded as not explainable in their raw form. From this starting point, two types of approaches emerge: methods that aim to reproduce the performance of deep learning systems without using deep learning itself, and methods that aim to make deep learning models themselves explainable.

In the first approach, the idea is that there is nothing to explain in a trained neural network. It is seen as a simple composition of matrix multiplications followed by nonlinear functions, optimized through gradient descent. There is no semantic meaning associated with each computational component, and therefore nothing intrinsic to interpret. This corresponds to the so-called ”black box” view of neural networks. Under this assumption, XAI consists of replacing as much of the deep learning system as possible with alternative AI methods.

In the second approach, it is argued that neural networks can be explained. The generalization ability of neural networks, cannot be explained solely by statistical correlations. A neural network does not simply memorize a set of examples and replay them. Instead, it exhibits a capacity for generalization that suggests the emergence of abstraction.

For the moment, both perspectives coexist under the umbrella of XAI, and because the field is still relatively young, neither approach has established clear dominance over the other. As a result, XAI is characterized by a wide diversity of methods and a high level of ongoing exploration. This richness reflects the fact that many possible directions are still open and actively being investigated.

Now that we have established a definition of what an explanation in XIA, we can investigate what this means in the context of reinforcement learning. In reinforcement learning, explaining refers to explaining the decision-making process of the agent. We assume that the structure of the environment as well as the agent's perception and action capabilities are known to the observer and do not need to be explained. These components are designed by humans and are therefore considered understandable by default.

The core element of interest is the agent's decision-making process, which is represented by the policy, denoted π. The policy is a function that maps observations $\mathcal{O}$ to actions $\mathcal{A}$. This policy is learned in order to maximize a reward function.

This setting makes the problem more difficult, since explaining a reinforcement learning policy is more challenging than explaining classical systems in XAI. In standard XAI settings, explanations often focus on a single decision problem, such as determining whether an image contains a specific object. However, reinforcement learning introduces a temporal dimension. This means that when explaining the choice of an action under a given policy, one must take into account a sequence of states and actions, which may involve uncertainty over time.